# Exercise series 5: SQL and Aggregation

Introduction to databases

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The employees database schema is defined as:

- **employees**(emp_no, birth date, first_name, last_name, gender,hire_date)
- **departments**(dept_no, dept_name)
- **dept_manager**(emp_no, dept_no, from_date, to_date)
- **dept_emp**(emp_no, dept_no, from_date, to_date)
- **titles**(emp_no, title, from_date, to_date)
- **salaries** (emp_no, salary, from_date, to_date)

- Note that most tables contain many (thousands or millions) of records, so using **LIMIT** is important.
- If you need to use the current date you can use the function **NOW()** that outputs the current date and time.
- You can use the **DROP** command to remove a table.
- You can use the **OR REPLACE VIEW** command to replace an existing view

```
SELECT * FROM departments LIMIT 10;

SELECT * FROM dept_emp WHERE to_date> NOW();

DROP TABLE test_table;

CREATE OR REPLACE VIEW test_view AS […];
```

# Write the following queries in SQL

1. What is the name of the last employee in the database?
2. List the department he works for?
3. What is the name of his current boss?
4. What is his current salary?
5. List the number of employees working in each department.
6. List all departments and the name of the current manager.
7. Find the names of managers that earn less than the highest paid manager.
8. Find all employees that have worked for at least two departments.
9. Find all employees that had the same title during their career.
10. ! Find all employees that had all possible titles during their career.
11. ! Find all employees that had an increase, between their first and current salary, that is at least 2 times the average increase in salary for any employee in the same department.
12. Find the top-10 employees with the highest last salary
13. Find for each department the total, maximal, average and minimal salary.

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1) What is the name of the last employee in the database?

```sql
SELECT first_name, last_name
FROM employees
ORDER BY emp_no DESC
LIMIT 1;
```

2) List the department he works for?

```sql
SELECT D.dept_name
FROM dept_emp DE, departments D
WHERE
	D.dept_no = DE.dept_no AND
	DE.emp_no IN
	(
		SELECT emp_no
		FROM employees
		ORDER BY emp_no DESC
		LIMIT 1
	);
```

3) What is the name of his current boss?

```sql
SELECT E.first_name, E.last_name
FROM dept_manager DM, employees E
WHERE
	DM.emp_no = E.emp_no AND
	DM.to_date >= now() AND
	DM.dept_no IN
	(
		SELECT dept_no
		FROM dept_emp
		WHERE
			to_date >= now() AND
			emp_no IN
			(
				SELECT emp_no
				FROM employees
				ORDER BY emp_no DESC
				LIMIT 1
			)
	);
```

4) What is his current salary?

```sql
SELECT salary
FROM salaries
WHERE
	to_date >= now() AND
	emp_no IN
	(
		SELECT emp_no
		FROM employees
		ORDER BY emp_no DESC
		LIMIT 1
	);
```

5) List the number of employees working in each department.

```
SELECT COUNT(distinct emp_no), dept_name
FROM dept_emp DE, departments D
WHERE
	to_date > now() AND
	DE.dept_no = D.dept_no
GROUP BY dept_name
;
```

6) List all departments and the name of the current manager.

```
SELECT first_name, last_name, dept_name
FROM dept_manager DM, departments D, Employees E
WHERE
		E.emp_no = DM.emp_no AND
		D.dept_no = DM.dept_no AND
		to_date > now()
;
```

7) Find the names of managers that earn less than the highest paid manager.

```
CREATE OR REPLACE VIEW manager_salaries AS
SELECT DM.emp_no AS emp_no, salary
FROM dept_manager DM, salaries S
WHERE
	DM.emp_no = S.emp_no AND
	S.to_date > now() AND
	DM.to_date > now()
;

SELECT first_name, last_name
FROM manager_salaries MS NATURAL JOIN employees E
WHERE EXISTS
(
	SELECT *
	FROM manager_salaries
	WHERE salary > MS.salary
);
```

8) Find all employees that have worked for at least two departments.

```
SELECT first_name, last_name
FROM employees
WHERE emp_no IN
(
	SELECT emp_no
	FROM dept_emp
	GROUP BY emp_no
	HAVING COUNT(distinct dept_no) > 1
)
LIMIT 100;
```

9) Find all employees that had the same title during their career.

```
SELECT first_name, last_name
FROM employees
WHERE emp_no IN
(
	SELECT emp_no
	FROM titles
	GROUP BY emp_no
	HAVING COUNT(distinct title) = 1
)
LIMIT 100;
```

10) Find all employees that had all possible titles during their career.

```
SELECT emp_no
FROM titles
GROUP BY emp_no
HAVING COUNT(distinct title) =
(
	SELECT COUNT(distinct title)
	FROM titles
);
```

## 11) Find all employees that had an increase, between their first and current salary, that is at least 2 times the average incre …

```
CREATE OR REPLACE VIEW currently_working_increase AS
SELECT S1.emp_no AS emp_no, S2.salary - S1.salary AS increase
FROM salaries S1 JOIN salaries S2 ON S1.emp_no = S2.emp_no
WHERE
	S2.to_date > now() AND
	(S1.emp_no, S1.from_date, S2.to_date) IN
	(
		SELECT emp_no, MIN(from_date), MAX(to_date)
		FROM salaries
		GROUP BY emp_no
	);

CREATE OR REPLACE VIEW aveg3 AS
SELECT 2*AVG(increase) AS aveg, dept_no
FROM currently_working_increase NATURAL JOIN dept_emp
WHERE to_date > now()
GROUP BY dept_no;

SELECT emp_no
FROM currently_working_increase CW NATURAL JOIN dept_emp NATURAL JOIN aveg3
WHERE
	to_date > now() AND
	increase > aveg;
```

## 12) Find the top-10 employees with the highest last salary

```
SELECT emp_no, salary
FROM salaries
WHERE (emp_no, to_date) IN
(
	SELECT emp_no, MAX(to_date)
	FROM salaries
	GROUP BY emp_no
)
ORDER BY salary DESC
LIMIT 10;
```

## 13) Find for each department the total, maximal, average and minimal salary.

```
SELECT dept_name, SUM(salary), MAX(salary), AVG(salary), MIN(salary)
FROM dept_emp DE, salaries S, departments D
WHERE
	DE.emp_no = S.emp_no AND
	D.dept_no = DE.dept_no
GROUP BY dept_name;
```